

MCS-1: Probability and Statistics (MA6.302)

Monsoon 2026

Lecture 4

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Definition: Conditional Independence

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$$P(A \cap B|C) = P(A|B \cap C)P(B|C) = P(A|C)P(B|C) \quad (1)$$

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- Let A be the event that the first urn was chosen, let R_1 be the event that the first ball was red, and let R_2 be the event that the second ball was red.
- Are R_1 and R_2 independent ?

Conditional Independence: Urn Example (continue)

Think about it intuitively first.

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- Then you know for sure that the first urn is picked.
- So the second ball has to be red as well.
- Knowing about the first ball tells you a lot about the color of the second ball.
- Clearly they are not independent

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- Let's compare: $P(R_1)P(R_2) = 0.5 \times 0.5 = 0.25$
- So, $P(R_1 \cap R_2) = P(R_1)P(R_2)$. Thus, $R_1 \not\perp R_2$.

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- **Conditional independence does not imply independence!**

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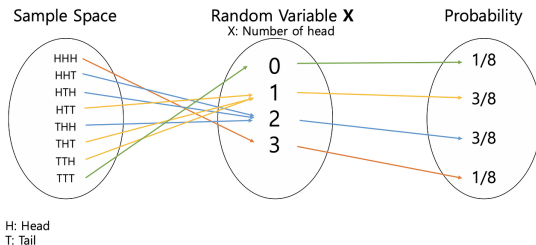
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- Conditioned E , you know that the tosses are different. Let the first toss is a H . Does this tell you anything about the second toss?
- Of course! The second toss has to be a T ! So intuitively, H_1 and H_2 should not be independent given E .

Toss 3 Coin Example



- Let X be the random variable associated with number of heads in three coin tosses.
- Here $\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$ and $X : \Omega \rightarrow \{0, 1, 2, 3\}$.
- Then, $X(HHH) = 3$, $X(HHT) = X(HTH) = X(THH) = 2$, $X(HTT) = X(THT) = X(TTH) = 1$ and $X(TTT) = 0$.

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- $X = \text{number of coin tosses to get a head. } X \in \{1, 2, 3, \dots\}$

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- A random variable is always written with upper-case and the numerical value we are trying to evaluate the probability for is written with a lower case.

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 - Well at least one head is $\{HT, TH, HH\}$. So probability of this is $3/4$.

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- For an uniform random variable X , each value has equal probability mass.
- If an uniform random variable X takes on k different values, then the probability mass at each of those values are $1/k$.

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 - It may rain one day or not.

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What if the coin is biased? Probability of H is 0.8?

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- So, $P(X = 5) = {}^{10}C_5 p^5(1 - p)^5$.
- What about $P(X = 8)$? $P(X = 8) = {}^{10}C_8 p^8(1 - p)^2$.

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- $P(Y = 0) = P(\{\text{no heads}\}) = (1 - p)^n$.

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- Let X_i be a Bernoulli random variable which is 1 if the i^{th} toss gave a head. Then, $\{X_i, i = 1, \dots, n\}$ are independent Bernoullis.
- Let Y be the number of heads we see at the end of all n tosses.
- Whats the relationship of Y to the X_i 's?

Y is called a Binomial random variable. What is the PMF of Y ?

- $P(Y = 0) = P(\{\text{no heads}\}) = (1 - p)^n$.
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- $\sum_{k=0}^n P(Y = k) = \sum_{k=0}^n {}^n C_k p^k (1 - p)^{n-k} = 1$ (**Binomial Expansion!**)

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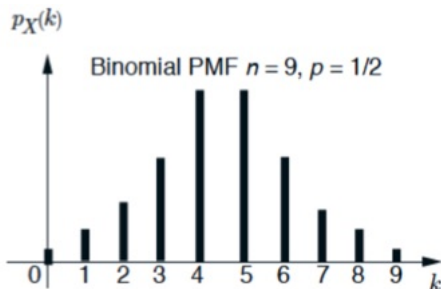
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- X is a *Binomial*(9, 3) random variable!

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- The number of trials required to get a single success is a **Geometric Random Variable**.

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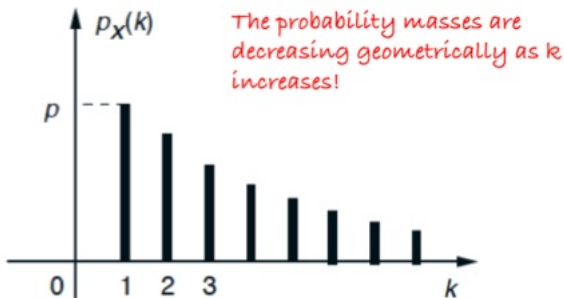
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- This probability is $P(X \geq k) = (1-p)^{k-1}$.
- $X > k$ is the event that $X \geq k+1$, and so $P(X > k) = (1-p)^k$.